Savyasanchi Aggarwal

aggarwalsavya@gmail.com | https://github.com/savya10 | https://www.linkedin.com/in/svyagg/

Education

University College London — *London, UK*

PhD Candidate, Computational Materials Science — *Sept 2022 - Sept 2026*

- Recipient of the **ARAP Research Scholarship** to study a joint PhD between UCL (with Professor David O. Scanlon) and A*STAR Research Institute, Singapore (with Dr Kedar Hippalgaonkar).
- Thesis title: *Accelerating materials design of defect-implemented semiconductors with machine learning*. (submitted September 2026)
- Work authorisation: UK citizen, US; requires sponsorship (O-1 eligible); **available to start from October 2026**.

Imperial College London — *London, UK*

MSci Chemistry: First Class Honours — *Oct 2018 - June 2022*

- **Relevant coursework:** Object-oriented programming, Density Functional Theory (DFT) for materials science (CRYSTAL17), Molecular Dynamics (VMD, LAMMPS, ASE), Solid-state electronics and magnetism modelling
- **Relevant courses:** Multivariate Calculus, Quantum Mechanics, Electromagnetism, Nanomaterials, Crystal Structures, Biological chemistry, Renewable & Energy Materials

The Judd School — *Tonbridge, UK*

A-Levels: A* Mathematics, A Chemistry, A Biology — *Sept 2010 - July 2017*

- 2017 Chemistry UK Olympiad: Gold
- 2016 C3L6 Chemistry Olympiad: Silver
- 2017 International Biology Olympiad: Special Mention

Experience

PhD Candidate in Computational Chemistry — *UCL/A*STAR Singapore*

Scanlon Materials Theory Group (Supervised by Prof. David O. Scanlon) — *Sept 2022 - Present*

- Developing bespoke ML models to expedite high throughput screening for defects and dopants in semiconducting materials, navigating complex energy landscapes using deep learning, with a specialization in **equivariant charge-aware graph neural networks**.
- Proficient in implementation, interpretation and augmentation of several machine learning methodologies; regression, **graph neural networks**, **transfer learning**, **multi-task/fidelity learning** and **active learning**. Presented works at major international conferences (**ICLR, MRS, AI4X**) with multiple first-author papers submitted/in preparation.
- Expertise in utilising DFT to understand chemical behaviour for a broad range of chemical systems, especially regarding defects and disorder in crystals.
- Constructed a **high-throughput, multi-modal** pipeline (*vasp2db, autotraj*) to extract material properties from hybrid-DFT calculations (VASP) to construct the largest publicly available computational defects database. Used by 20+ users in the research group, connecting ARCHER2 National Supercomputing HPC cluster to a MongoDB noSQL database, hosted on a **cloud-based** server (Kamatera).
- Extensive experience in using High-Performance Computing (**HPC**) clusters (Tiers 1 to 3), creating tailored high-throughput workflows for automated submission/parsing/analysis (using DOPED, AiiDA, atomate2) with slurm and SQE queue systems.
- As the only researcher in the group with a focus on machine learning (2025), self taught and conceptualised multiple research projects, helping secure a £2m funding grant for my supervisor. Directly supervised 3+ other PhD students and assisted with other projects.
- Active co-developer/contributor to several open-source packages, including ***vasp2db***, ***autotraj***, ***DOPED***, ***ShakeNBreak***, **defectivator**, **mlippr**
- **Technical Skills:** Python, Deep Learning (GNNs, Multi-task learning, Regression/Classification, Dimensionality Reduction, LLMs), Data Augmentation, High-throughput workflows (*AiiDA, atomate2*), Hybrid-DFT (VASP), HPC cluster usage (SQE/slurm parallelisation), MongoDB, Linux, Bash scripting

MSci Chemistry Research Student — *Imperial College London*

Harrison Computational Materials Group (Supervised by Dr. Giuseppe Mallia & Prof. Nic Harrison) — *Oct 2021 - Jul 2022*

- Investigated the role of defects in integrating low-dimensional materials into modern capacitor technology, focusing on 2D silicon (silicene).
- Used Gaussian-based Hybrid-DFT to model electronic and structural properties of silicene interfaced with larger electronic components.
- **Technical Skills:** Density Functional Theory with CRYSTAL17, Python, Linux, HPC cluster usage, Bash scripting

Undergraduate Researcher - Computational Materials Science *Imperial College London*

Materials Design Group (Supervised by Prof. Aron Walsh) *Jun 2021 - Feb 2024*

- Lead developer and maintainer of an **open-source Python library**, *PyTASER*, to predict transient-absorption-spectroscopy from first principles.
- Designed the package to rapidly classify and analyse experimental spectra from local DFT data, leveraging pythonic architecture to isolate individual band transitions from generated spectra (*beyond experimental capability*).
- **First-authored paper** published in the *Journal of Open Source Software* (JOSS), with another follow-up theoretical paper in preparation.
- Supervised an intern to extend the project, and onboarded multiple PhD-level researchers upon their joining the project (2022)
- **Technical Skills:** Object-oriented scientific programming with Python, Git, Website development with Sphinx and ReadtheDocs

Undergraduate Researcher - Physical Chemistry *Imperial College London*

Kuimova Research Group (Supervised by Dr. Marina Kuimova) *Aug 2021 - Oct 2021*

- Synthesised and tested affinity of different optical probes against the lifetime of DNA G-quadruplex bindings, with the goal of improving **biosensor technology**, performing analysis with laser fluorescence (among other) spectroscopy. Co-authored paper accepted in *RSC Analytical Chemistry*.

Research & Development Intern - Chemistry *London, UK*

NOTPLA Bioplastics *Jul 2019 - Sept 2019*

- Tested different manufacturing and extraction methods for the development of biodegradable single-use plastics in a rising bioplastics startup.
- The work was implemented directly into the industrial workflow, playing a key role in securing a **£10m seed fund** and the **Earthshot Prize 2022**

Publications

Published

S.Aggarwal, Y.W. Woo, S.R. Kavanagh, A.Ganose, A.Walsh, ***PyTASER*: A tool to simulate transient absorption spectroscopy (TAS) from first principles**, *Journal of Open Source Software* (2024). doi.org/10.21105/joss.05999

T.Bradford, P.A Summers, Y.L.J. Lam, **S.Aggarwal**, J.B. Vannier, R. Vilar, M.Kuimova, **The effectiveness of TO and ThT as optical probes for G4 investigations**, *Analytical Chemistry* (2024). doi.org/10.1021/acs.analchem.4c04207

A.Roy, ..., **S.Aggarwal**, et al., **Outcomes of the 2025 Large Language Model (LLM) Hackathon for Applications in Materials Science and Chemistry**, *arXiv preprint* (2025). arXiv:2605.03205

Under review & peer-reviewed workshop papers

M.H. Petersen, R. Zhu, H. Dai, **S.Aggarwal**, W.Nong, A.P.Chen, A.Bhowmik, J.M.Garcia-Lastra, K.Hippalgaonkar, **Dis-GEN: Disordered crystal structures** — under review, *npj Comp. Mater.*; **accepted at ICLR 2025 (AI4Mat Workshop)**

S.Aggarwal, M.H. Petersen, S.R. Kavanagh, A.G. Squires, K.Hippalgaonkar, D.O.Scanlon, **Predicting formation energies for charged defects in semiconductors using machine learning — accepted at AI4X 2025**; journal manuscript in preparation

In preparation

S.Aggarwal, C. Parida, P.P.Russell, M.H.Petersen, R. Claes, S.R. Kavanagh, A.G. Squires, K.Hippalgaonkar, D.O.Scanlon, **Machine learning defect thermodynamics for high-throughput dopability screening** — targeting *npj Comp. Mater.*, submission Oct 2026

P.P. Russell, **S.Aggarwal**, R. Claes, A.G. Squires, D.O.Scanlon, **Leveraging machine learning for discovery of metastable charged defects** — targeting *npj Comp. Mater.*, submission Oct 2026

S.Aggarwal, S.R. Kavanagh, D.O.Scanlon, **Valence alternation defect behaviour in II-VI compounds** — targeting *Chem. Mater.*, submission Oct 2026

Y.W. Woo, S.R. Kavanagh, **S.Aggarwal**, A.Ganose, A.Walsh, **Transient Absorption Spectroscopy of Crystalline Semiconductors from First-Principles** — targeting *JACS*

Conferences

Talks

- *2026 Materials Research Society International Conference,* ***Honolulu, Hawai'i***
- *2026 Thomas Young Centre Postgraduate Day,* ***King's College London***
- *2025 AI4X International Conference for Applied Machine Learning,* ***National University of Singapore***
- *2025 Materials Research Society International Conference Spring,* ***Seattle, US***
- *2025 PGR Symposium for Computational Chemistry,* ***University College London***
- *2024 MCC Workshop on the Modelling of Point Defects,* ***Lancaster University***
- *2023 Materials Chemistry Consortium,* ***Daresbury Laboratory***
- *2023 Solid State Group Meeting for Early Careers Researchers, Virtual* - Received ***"Best Oral Presentation"*** *prize*

Posters

- *2025 Solid State Group Meeting,* ***Leeds University***
- *2025 ICLR AI4Mat Workshop,* ***Singapore***
- *2024 Solid State Group Meeting,* ***Birmingham University***
- *2023 TYC Postgraduate Day,* ***University College London***
- *2022 Solid State Group Meeting,* ***Cambridge University***

Leadership and Activities

LLMs for Materials Hackathon 2025 *2025*

Winner of 2025 Visionary Award

- Led a group of 6 PhD/postdocs to develop a LLM model (DFTPilot) to easily translate experimental setups to DFT simulation frameworks.
- Winner of the ***2025 Visionary Award***, for top performing teams out of 120 total submitted projects (1200+ participants).
- Use of Graph NNs, RAG for developing backend and Fast API for front-end. Currently expanding this work for use in internal pipelines.

Technology Development/Entrepreneurship *2019 - Present*

- **FoNS CSP Datathon 2021**: Led a group of five to optimise regression and classification models for predicting crystal structures from the CCDS dataset of 13,000+ molecular structures and 2000+ features. Secured runner-up with a maximum performance of 88%.
- **FoNS-MaD Entrepreneurship Competition 2019-2020**: Developed a biosensor system to allow rapid & cheap detection of water-borne pathogens for use in the third-world. Worked with Si/Au nanoparticle-fixed aptamer based technology, backed by leading professors in biosensing and reached the penultimate stage of the competition.
- **E.Wash 2019-2020**: Worked closely with a student organisation (e.quinox society) to develop a sustainable off-grid washing machine for use in rural Rwanda, hoping to encourage systemic societal change through development of grassroot-empowering technology.

Imperial College Taekwondo Society *Imperial College London*

Union President (2020-22), Secretary (2019-20) *2019-22*

- Led a society with an annual membership of 100+ members for 2 academic years.
- Organised and taught regular training sessions, directly handling budgeting and negotiations with external companies and the student union.
- Directly handled the transition towards and away from COVID-19 lockdown, maintaining membership and morale in the society.

Private Tutor *2016-Present*

- Tutor for university-level (Chemistry) and high-school students (Mathematics, Chemistry, Biology).
- Tutored undergraduate students to assist their application process for high-ranking US & UK-based Masters/PhD postgraduate programs. 100% success rate (9 students), with all students achieving desired schools (Harvard, Stanford, Cambridge, Oxford, UCB, ...).

Technical Skills

Machine Learning	Regression, GNNs & MPNNs, Transfer Learning, Multi-task models, Generative Models (VAE, Diffusion Models), LLMs
Scientific Computing	*Python* (OOP, Pytorch, ScikitLearn, Tensorflow), HPC Cluster usage (slurm & SQE), MongoDB, Git, vim, VASP (& analysis/implementation packages), Shell (Bash/Zsh)

Interests

Taekwondo	ITF Black belt 1st Dan, competing and coaching for 15+ years. *International-level competitor representing England (-75kg).*
Chess	ECF ranked national-level player (5+ years, 2008-2013).
Languages	I am keen to learn new languages. Currently known languages: English (native), Hindi (native), Spanish (conversational), Cantonese (amateur), French (amateur)